



36 levels, 92 lines. Every number is tabulated in `data/transitions.csv`

A, B: magnetic-dipole and electric-quadrupole constants as compiled in ARC.

$\lambda$  and  $\nu$  from NIST level energies (centre of gravity, no hyperfine structure);  $\lambda_{\text{air}}$  uses standard air (Ciddor). A: NIST-compiled value where available, otherwise ARC.  $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ . Branch: share of the upper level's decays that takes this line.

Rydberg term energies from pairinteraction (quantum defects), lower level from NIST.  $d_{70} = |\langle J||er||J' \rangle|$  to  $n = 70$  in  $ea_0$ , from pairinteraction and from ARC. The two disagree by a factor  $\approx 2$  for  $5S \rightarrow nP$  (near-cancelling integral): treat that row as an order of magnitude.